

Agents

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What is an Agent?

- A Generative AI agent is an application that attempts to achieve a goal by observing the world and acting upon it using the tools that it has at its disposal.
- Agents are **autonomous** and can act independently of human intervention.
- Even in the absence of explicit instruction sets from a human, an agent can reason about what it should do next to achieve its ultimate goal.

Expense Agent

What can I help you with?

Can I claim my monthly cellphone bill as an expense?

Yes, you can claim up to \$75/month. I can automatically submit this expense for you each month if you'd like.

Type your message here...

Send

Travel Agent

What can I help you with?

I need to book a flight to Seattle for next week's conference.

I've found a flight for \$450 and booked a hotel. I'll automatically submit the expense claim with receipts for you.

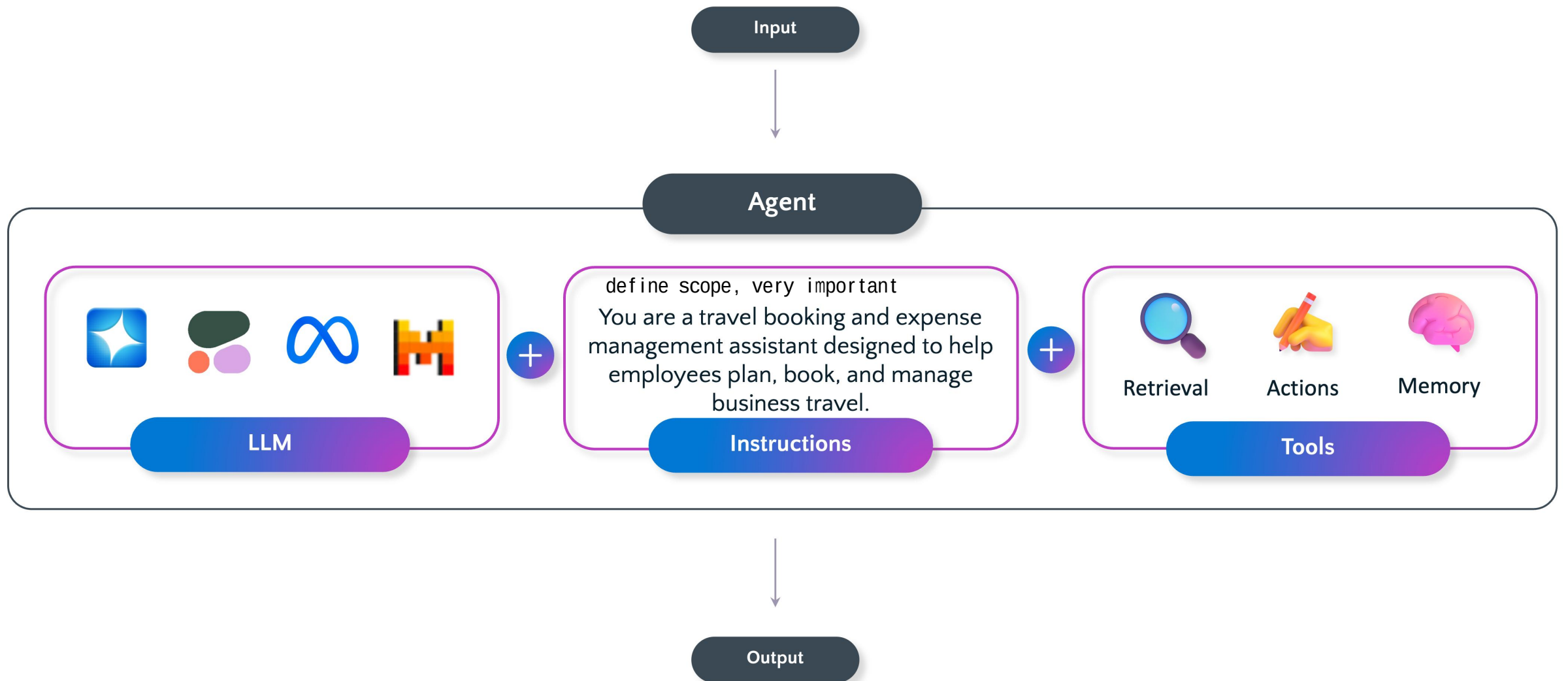
Type your message here...

Send

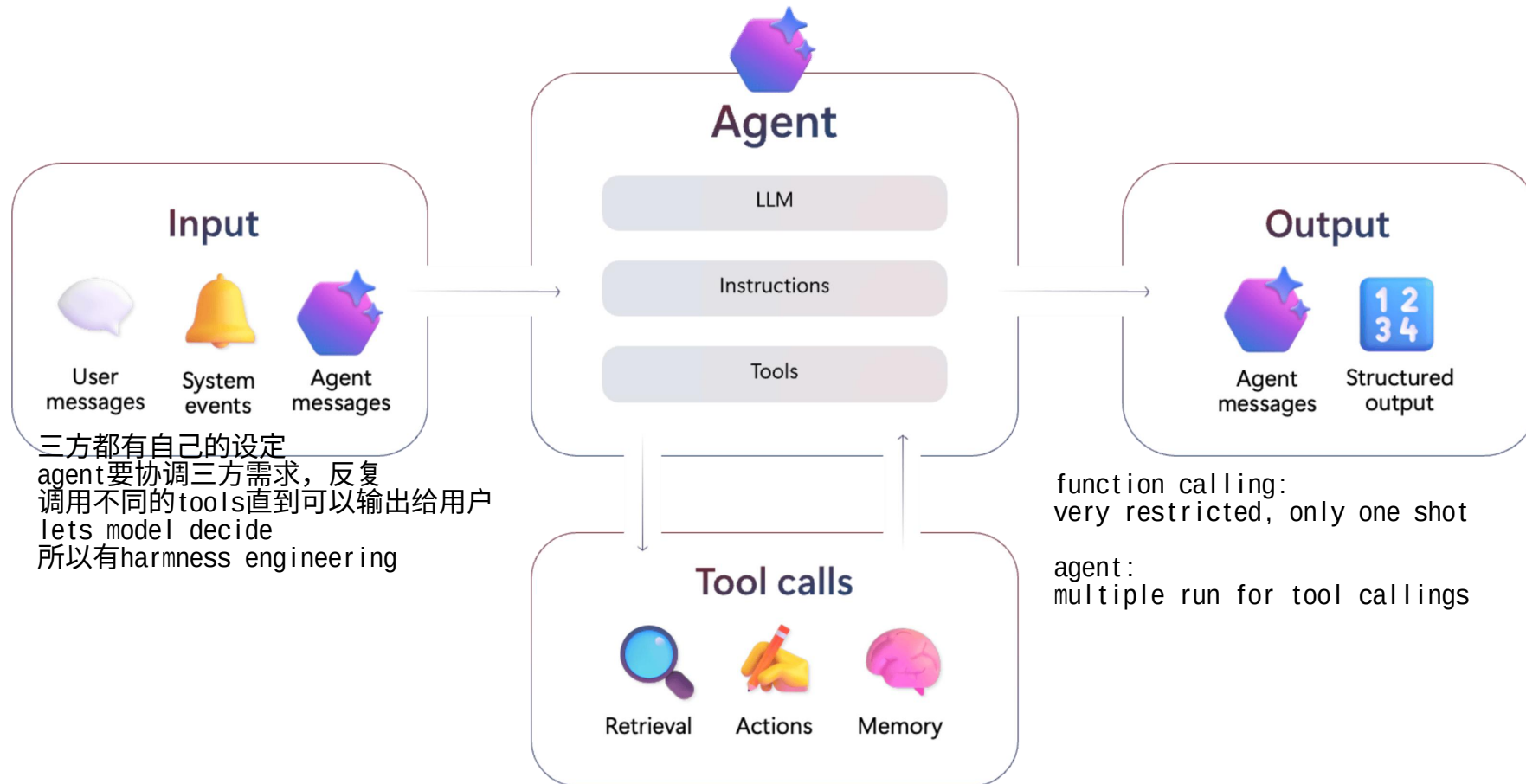
Model vs. Agent

Aspect	Model	Agent
Knowledge Scope	Limited to training data	Extended through external systems via tools
Inference and Prediction	Single inference based on user query, no management of session history	Managed session history for multi-turn inference
Tool Implementation	No native tool implementation	Tools natively implemented
Logic Layer	No native logic layer, relies on user prompts	Native cognitive architecture with reasoning frameworks like CoT and ReAct

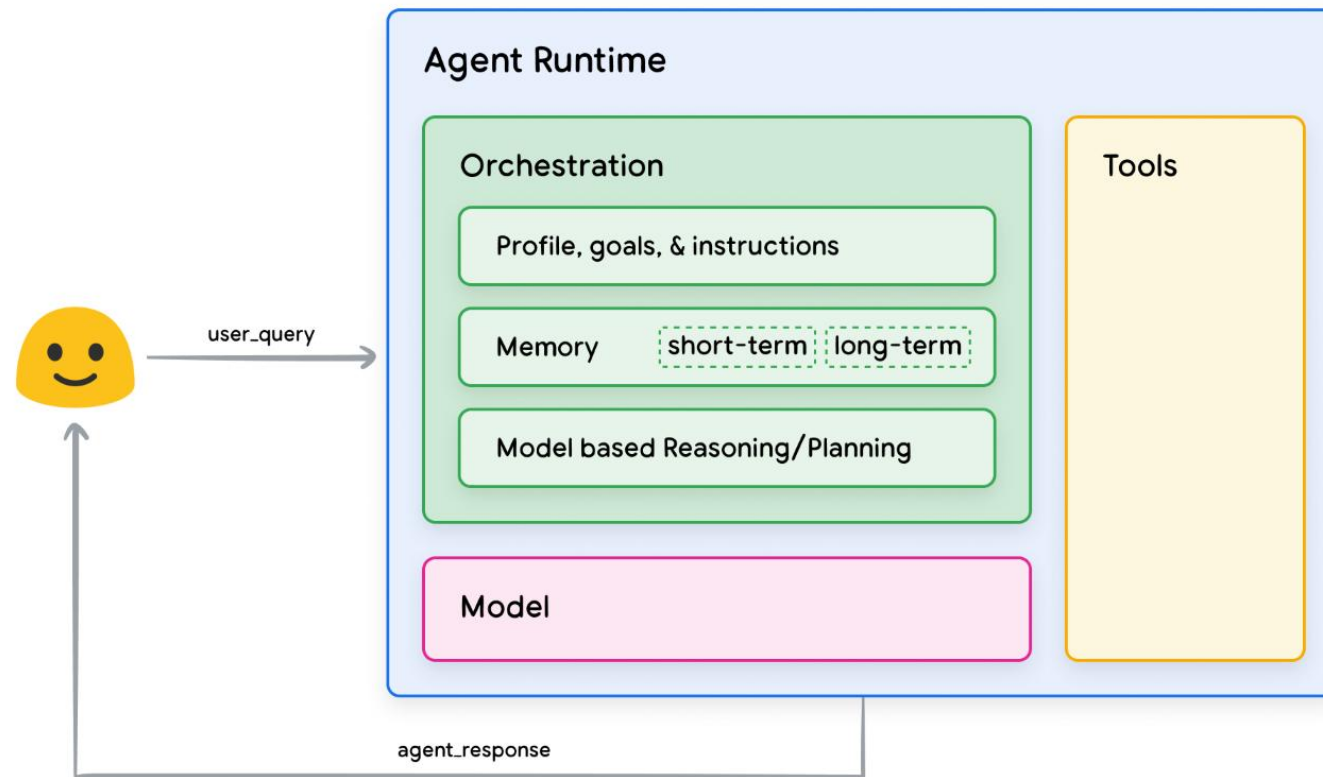
Agent



Agent

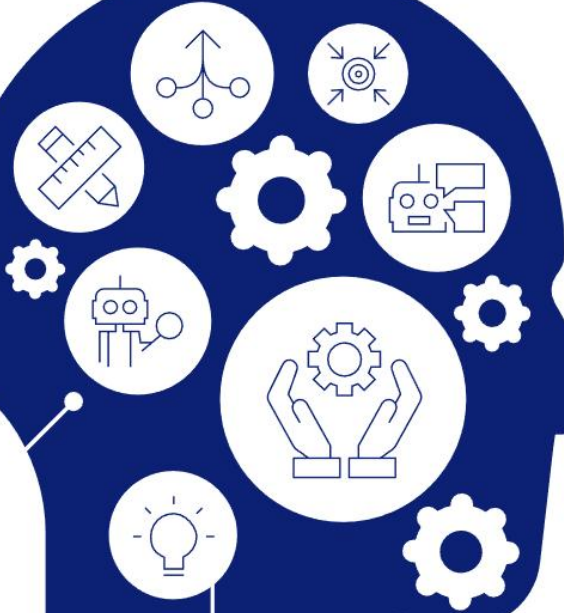


Agent Architecture and Components



Autonomy

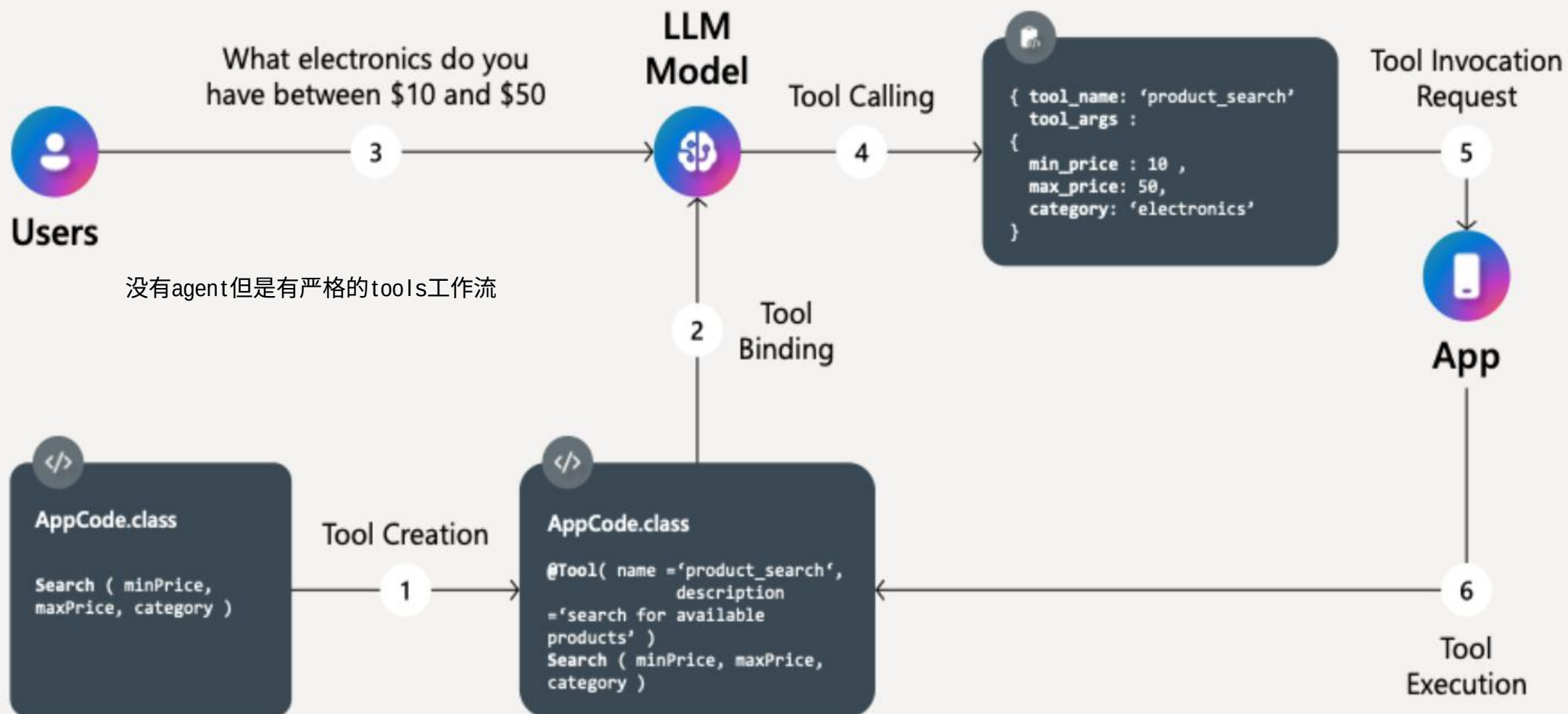
“Agency”



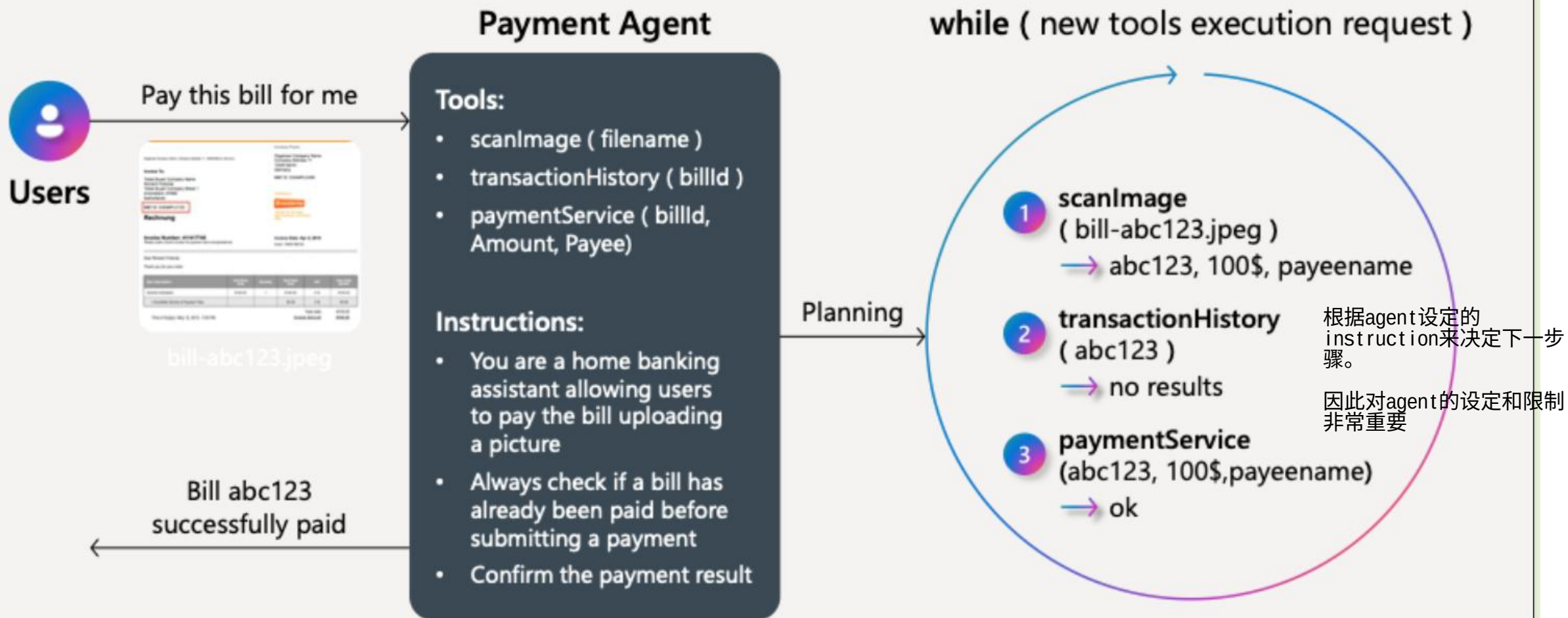
- multiple agent

McKinsey & Company

Agentic Pattern - Tools Calling



Agentic Pattern - ReAct Planning with Tools Calling



important非常重要的设置instruction, 展现作为人类的reasoning process

ReAct

- Combines reasoning and acting for better decision-making.
- Enhances LLMs by integrating logical reasoning with actions.
- Improves response accuracy and relevance.
- Two Components:
 - Reasoning: Logical inference and deduction.
 - Acting: Executing actions based on reasoning.

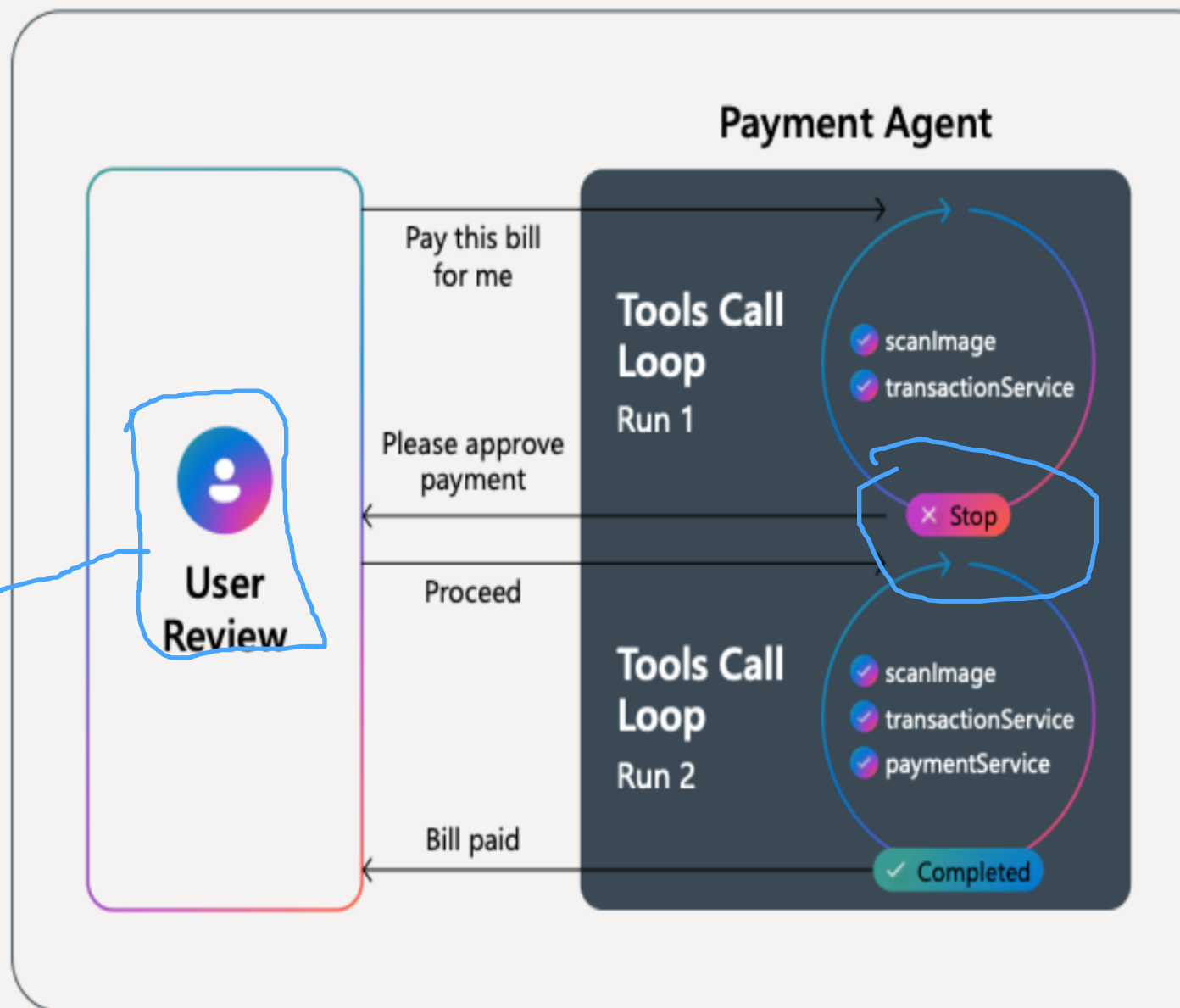
Agentic Pattern - Flow control

Looping Termination

- MaxIterations 最大迭代次数
- Message termination
- Human step /Human in the loop

Human in the loop

- Action execution approval
- Escalation
- Data review



Tools

- Function calling
- Code interpreter
- File search (simple RAG with user-uploaded files)
- Azure AI search (existing AI search index)
- Web search
- Grounding with Bing Search
- Grounding with Bing Custom Search (selected web domains)
- Model Context Protocol (MCP)

Tool support by region and model

- <https://learn.microsoft.com/en-us/azure/foundry/agents/concepts/tool-best-practice#tool-support-by-region-and-model>
- Code Interpreter NOT available in japaneast, southcentralus, southeastasia, spaincentra
- Computer Use ONLY available in eastus2, southindia, swedencentral
- Function NOT available in brazilsouth, northcentralus, southcentralus, westus

Agent: Function Calling

- Programs:
 - `agent_function_calling.py`
 - `agent_function_calling_v2.py`
- Create and use an agent with three functions

DefaultAzureCredential() | AzureCliCredential()

- Register on Azure Portal with your HKU email address
- Install Azure Command-Line Interface (CLI) on your computer
 - <https://learn.microsoft.com/en-us/cli/azure/?view=azure-cli-latest>
- Run az login to login with your hku email address and select the subscription associated with the Foundry project (**see next slide**)
 - <https://learn.microsoft.com/en-us/cli/azure/get-started-with-azure-cli?view=azure-cli-latest>
- Update the four environment variables starting with “FOUNDRY_PROJECT_” in the .env file based on the Foundry project
 - FOUNDRY_PROJECT_ENDPOINT=https://applied-llm-resource.services.ai.azure.com/api/projects/**tai-man-chan-project**

Select 3 (Subscription name=Teaching) when running az login

```
4.function calling % az login
A web browser has been opened at https://login.microsoftonline.com/organization
s/oauth2/v2.0/authorize. Please continue the login in the web browser. If no we
b browser is available or if the web browser fails to open, use device code flo
w with `az login --use-device-code`.

Retrieving tenants and subscriptions for the selection...

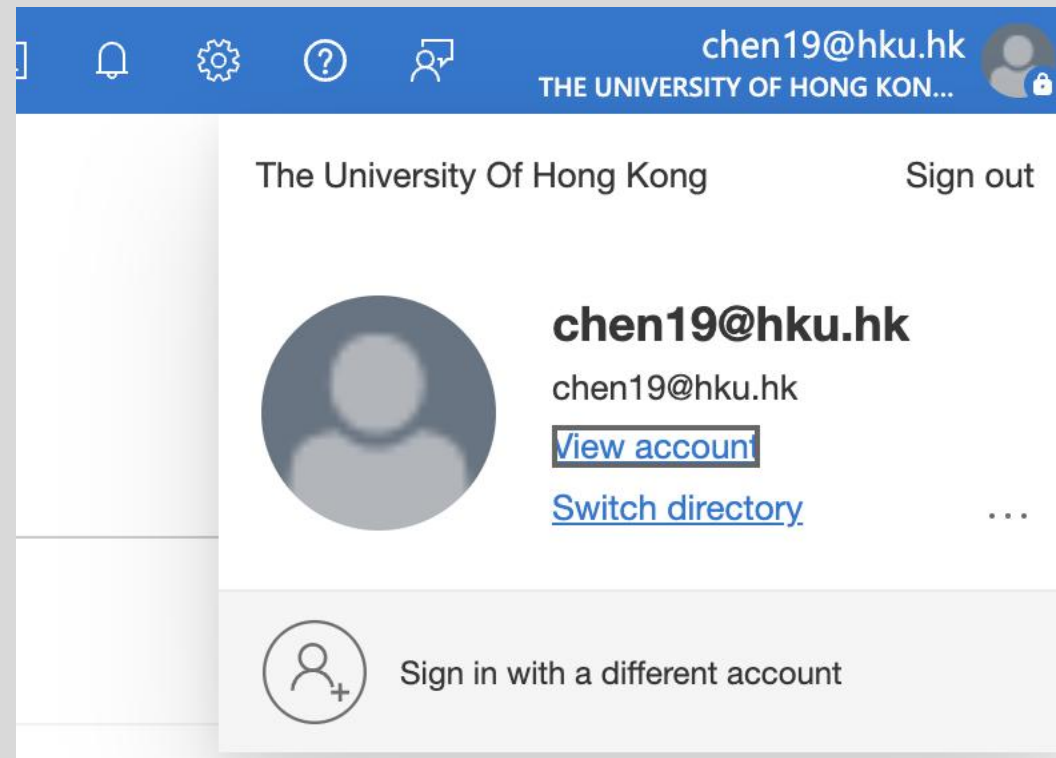
[Tenant and subscription selection]

No      Subscription name      Subscription ID      Tenant
-----
[1] *   Azure for Students  1d1d255f-[REDACTED]  The Universi
ty of Hong Kong - Connect
[2]     Azure subscription 1 b2af5377-[REDACTED]  The Universi
ty of Hong Kong - Connect
[3]     Teaching           425aa05e-881e-4108-9c49-57e55f4b2271  The Universi
ty Of Hong Kong

The default is marked with an *; the default tenant is 'The University of Hong
Kong - Connect' and subscription is 'Azure for Students' ([REDACTED]).

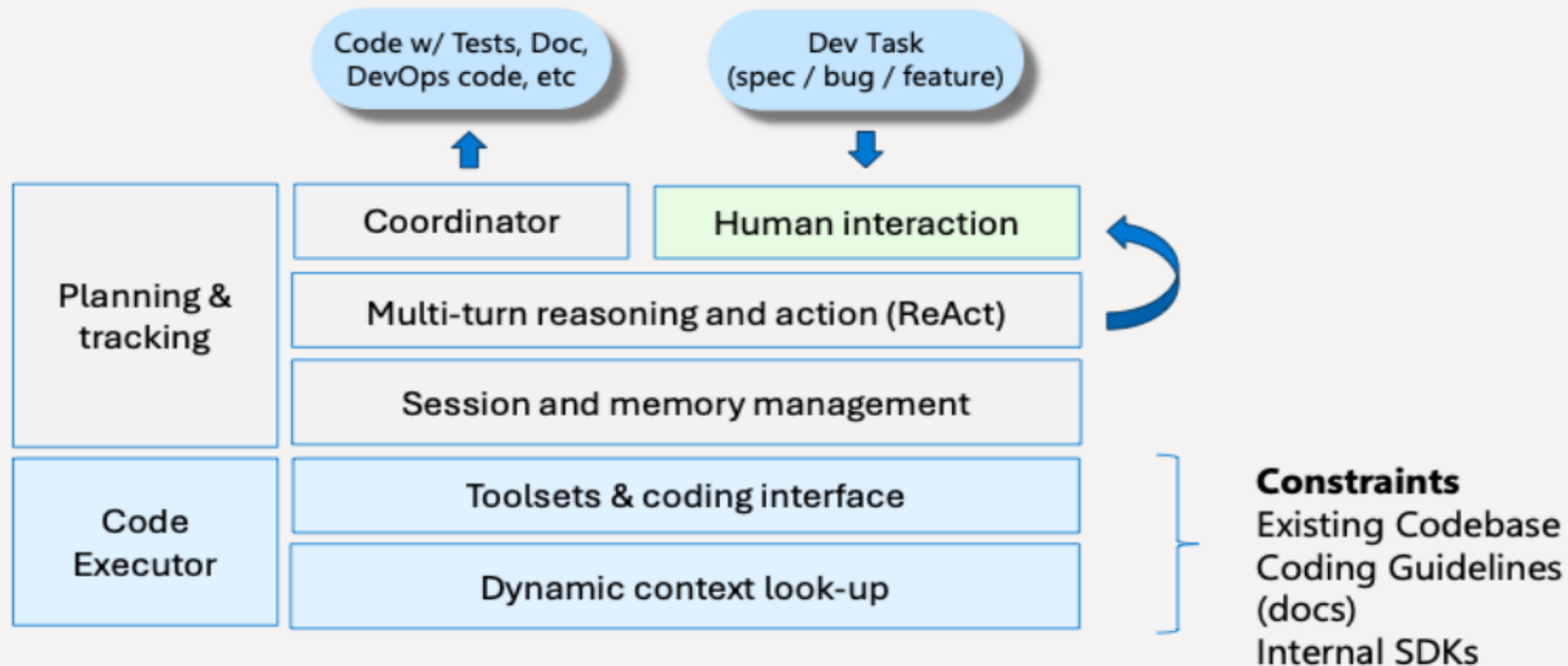
Select a subscription and tenant (Type a number or Enter for no changes):
```

Switch Directory in Azure



Code Generation Agent

Generates code based on natural language requirements, leveraging existing code base, templates, guidelines, libraries to match policies and best practices while interacting with humans to clarify, validate and deliver functionality as intended.



Agent with Code Interpreter

- Programs: `agent_code_interpreter_airbnb.py` (simple version) and `agent_code_interpreter.py` (more streamlined)
- Create an agent with Code Interpreter
- Use an existing agent for data analysis on your project

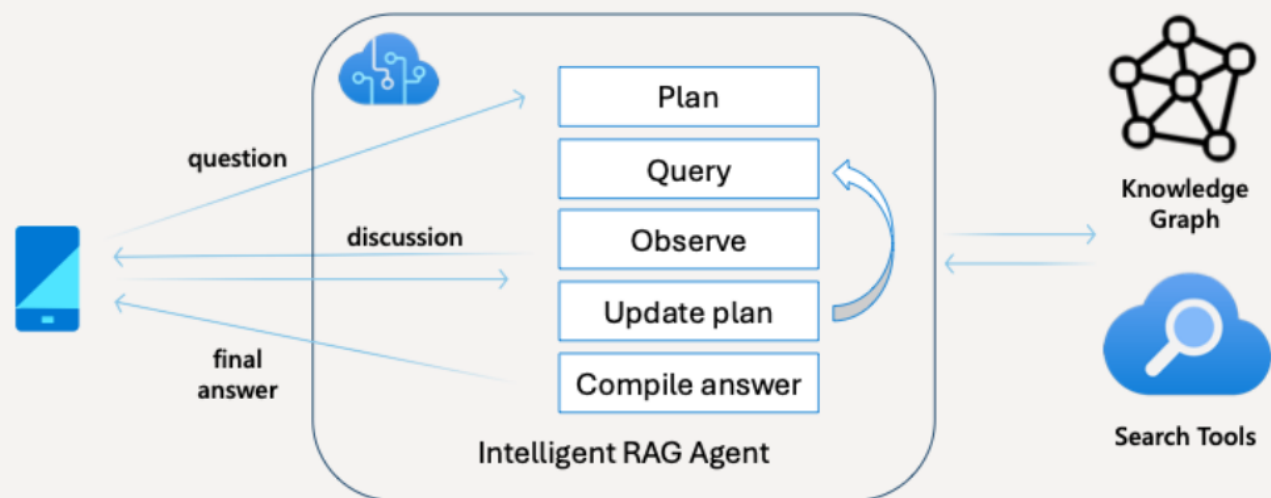
Code Interpreter 是大模型（LLM）的一种 Tool/Function Calling 能力，允许 AI 在对话中真正执行 Python 代码并返回结果。

Case Study: Movie DB Agent

- Agent Capabilities
 - Function Tool
 - Code Interpreter Tool
- Building Blocks / Code Examples:
 - `agent_function_calling.py`
 - `agent_code_interpreter.py`
 - `function_call_db.py` (from Lecture 4)
- Programs written by GitHub Copilot
 - `agent_function_call_db.py`
 - `agent_function_call_db_streamlit.py`

Retrieval Augmented Generation Agent

Translates questions into a research problem with human in the loop to produce high quality answers to complex questions within the scope of its domain



RAG Agent with File Search

- Programs
 - `agent_file_search.py`: create a new agent by uploading a file
 - `agent_file_search2.py`: use an existing agent
- allm-course-agent based on data/Course Introduction.pptx

allm-course-agent Web Interface

Microsoft Foundry / applied-llm

New Foundry Home Discover Build Operate Docs

allm-course-agent

v2 saved Today 10:09 am Save Preview Publish

Playground Traces Monitor Evaluation

Model: gpt-4.1-mini

Voice mode Preview
Switch the agent to a voice-first experience.

Instructions
You are a Course Assistant Agent for the Applied LLMs course.
Your role is to help students understand the course introduction, learning objectives, structure, assessment,

Tools
Connect tools to your agent for faster access to key information and improved efficiency. Learn more about best practices.
File search
index_allm_course • 22.49 KB
vs_ls5VaeqklhTXuSbW8WjGCq9X

Add

Chat YAML Code

7717t Metrics

give me an overview of the group project

Here is an overview of the group project for the Applied LLMs course:

- **Weight and Components:** The group project counts for 50% of the grade. It consists of:
 - Project Proposal (one page) worth 5%
 - Data and Program Code worth 15%
 - Project Report (up to 10 pages) worth 10%
 - Group Presentation worth 20%
- **Group Formation and Size:** Groups consist of 8 members each. Students initially find their own teams; otherwise, they will be assigned randomly. Leadership can rotate among members.
- **Project Ideas:** Some example topics are:
 1. AI-generated Financial Analyst Reports – generating analyst reports to support stock investment decisions using data like financial statements, Yahoo Finance, etc.
 2. Competitive Intelligence for Master Programs – collecting and analyzing public data on business school master programs to aid strategic decisions

Message the agent...

AI-generated content may be incorrect

RAG Agent with Web Search

- Foundry Agents
 - web-search-agent: `agent_web_search.py`
 - bing-search-agent
 - bing-custom-search-agent

Foundry IQ



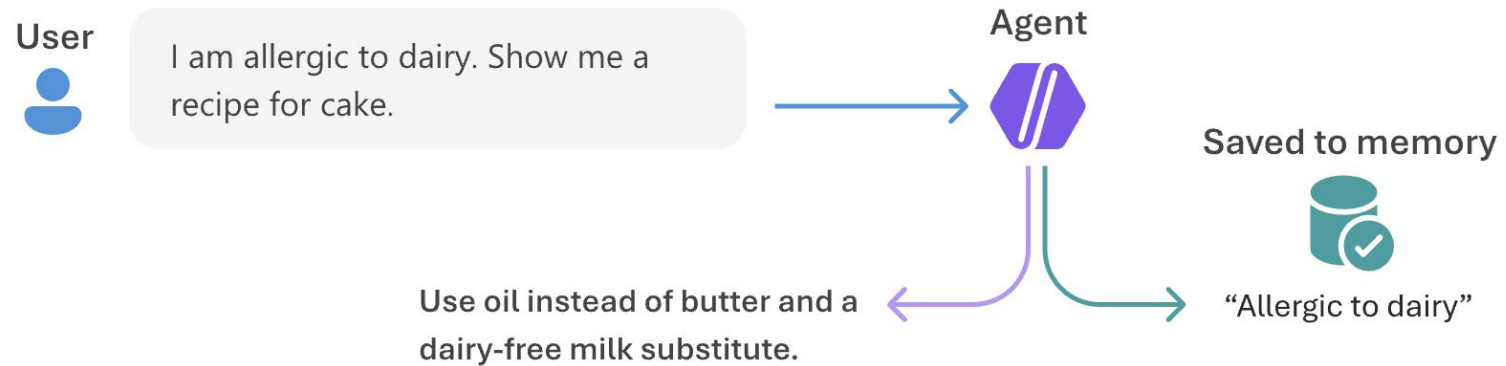
Agent with Foundry IQ

- Azure AI Search service
- Azure Storage account (Azure Blob Storage)
- Program: `agent_foundry_iq.py`

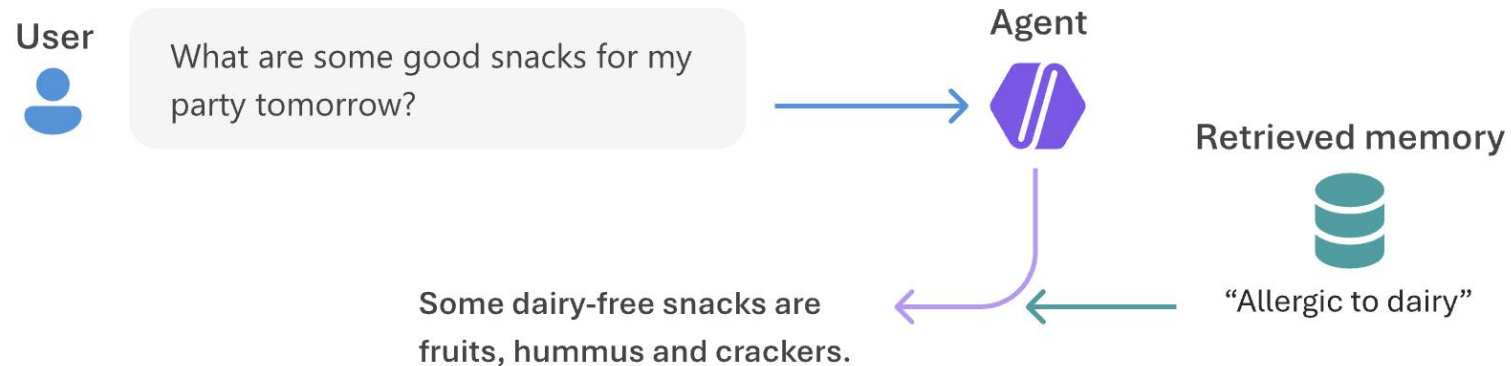


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How Memory Works



A few weeks later



Two Types of Long-term Memory

Type	Description
User profile memory	Information and preferences about the user, such as preferred name, dietary restrictions, and language preference. These memories are considered "static" with respect to a conversation because they generally don't depend on the current chat context. Retrieve user profile memories once at the beginning of each conversation.
Chat summary memory	A distilled summary of each topic or thread covered in a chat session. These memories allow users to continue conversations or reference prior sessions without repeating earlier context. Retrieve chat summary memories based on the current conversation to surface relevant threads.

Memory Use Cases

- A customer support agent that remembers your name, previous issues and resolutions, ticket numbers, and your preferred contact method (chat, email, or call back). This memory helps you avoid repeating information, so conversations are more efficient and satisfying.
- A personal shopping assistant that remembers your size in specific brands, preferred colors, past returns, and recent purchases. The agent can suggest relevant items as soon as you start a session and avoid recommending products you already own.
- A travel agent that knows your flight preferences (window or aisle), seat selections, food choices, nonstop versus connecting flights, loyalty programs, and feedback from past trips. The agent uses this information to quickly build an optimized itinerary.



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Security Risks of AI Agents

What you might experience	Risk area	What's happening
"The agent just shared confidential salary data in a customer chat!"	Data leakage and privacy exposure	The agent accessed sensitive information but lacked proper controls to prevent exposing it externally.
"Someone tricked the agent into revealing our database password."	Prompt injection and manipulation attacks	A malicious user crafted an input that overrode the agent's intended behavior.
"Our support agent is now deleting customer records—but it shouldn't have that permission!"	Unauthorized access and privilege escalation	Weak access controls allowed the agent to perform actions beyond its intended scope.
"The agent started recommending fraudulent products after we updated the training data."	Data poisoning	Someone corrupted the agent's training or contextual data, causing unsafe outputs.

Security Risks of AI Agents

What you might experience	Risk area	What's happening
"A third-party plugin we integrated is now sending our data to an unknown server."	Supply chain vulnerabilities	External dependencies introduced security vulnerabilities into the agent's workflow.
"The agent automatically processed a refund without verifying the request."	Over-reliance on autonomous actions	The agent executed an action without proper validation or human oversight.
"We can't figure out who accessed what data or when."	Inadequate auditability and logging	Missing or incomplete logs make it impossible to trace agent actions or detect misuse.
"Someone extracted customer information by repeatedly querying the agent."	Model inversion and output leakage	The attacker exploited model outputs to infer sensitive data from training or prompting.

Security Best Practices

Security Practice	What to Do	Why It Matters
Tight Access Control	Enforce role- based access control (RBAC) and least- privilege permissions	Limits damage if an agent is compromised
Input Validation	Apply prompt filtering and validation to all inputs	Prevents prompt injection and malicious instructions
Human Oversight	Gate high- risk or sensitive actions with human- in- the- loop approval	Avoids autonomous high- stakes mistakes
Comprehensive Logging	Log all agent actions with timestamps, actors, and rationale	Enables auditing, debugging, and accountability
Supply Chain Monitoring	Regularly audit third- party APIs, plugins, and tools	External dependencies are common attack vectors
Model Maintenance	Retrain and validate models to detect drift or poisoning	Maintains reliability and prevents silent degradation